

MACE: Multi-core Agentic Co-design Engine

Overview

We propose **MACE**, an agentic AI system for autonomous multicore hardware construction and verification, built on the **CHIA** framework. The project will be developed in two phases.

Phase 1 will extend CHIA with support for **OpenPiton**, enabling CHIA agents to configure, build, simulate, execute workloads, and collect results from the platform through a unified interface.

Phase 2 will use this extended CHIA infrastructure to build MACE: a multi-agent workflow that takes a hardware objective and autonomously decomposes it into engineering tasks, modifies the design, runs verification, analyzes failures, and iterates toward a working multicore system. Specialized agents will work in parallel where possible, with verification performed after each incremental change.

The project will demonstrate agentic capabilities on a realistic, large-scale open-source hardware platform rather than restricting the agents to isolated RTL generation.

Methodology

The first phase will implement an OpenPiton platform adapter following CHIA's existing platform abstractions. The adapter will expose the necessary build, simulation, workload execution, and result-collection operations to CHIA.

The second phase will introduce MACE's agentic workflow:

**Requirement → Planning → Task Decomposition →
Parallel Agent Execution**

**→ RTL/System Integration → Verification → Failure
Analysis → Iteration**

A planning agent will decompose system-level objectives into smaller tasks. Specialized agents will perform implementation and verification tasks through CHIA's tools. Each modification will be compiled and tested before subsequent tasks depend on it. Compilation or simulation failures will be fed back to the agents for diagnosis and correction.

The final demonstration will evaluate MACE on a multicore construction or integration task using OpenPiton, measuring both the correctness of the resulting system and the effectiveness of the autonomous workflow.

Expected Results

We expect to deliver:

- CHIA support for OpenPiton as a usable hardware platform.
- A working multi-agent MACE prototype built on the extended CHIA framework.
- Automated decomposition, implementation, verification, and debugging of multicore hardware tasks.
- Demonstration of parallel specialized agents and incremental verification.
- Quantitative evaluation of successful tasks, iterations, failures recovered, execution time, and compute usage.
- An open-source, reproducible demonstration of agentic hardware engineering.

Cost Estimate

We request **\$1000 USD** in compute credits. These resources will primarily support the computationally intensive experiments required for repeated RTL compilation, simulation, regression testing, and multicore workload evaluation.

AI inference for MACE will primarily use the AI/API credits provided through the CHIA program. Open-source EDA and simulation tools will be used wherever practical.